

Assignment No.11

Problem Statement : Implement Map reduces operation with suitable example using MongoDB.

Program :

```
show dbs
use customer
db.cust_table.insert({Item_id:1,Cust_Name:"Ram",Product:"Milk",Amount:40});
db.cust_table.insert({Item_id:2,Cust_Name:"Ram",Product:"Parle_G",Amount:50});
db.cust_table.insert({Item_id:3,Cust_Name:"Mohan",Product:"Lays Chips",Amount:40});
db.cust_table.insert({Item_id:4,Cust_Name:"Shivam",Product:"Mentos",Amount:10});
db.cust_table.insert({Item_id:5,Cust_Name:"Mohan",Product:"Maggie",Amount:60});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$sum:"$Amount"}}});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$avg:"$Amount"}}});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$min:"$Amount"}}});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$max:"$Amount"}}});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$first:"$Amount"}}});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$last:"$Amount"}}});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$psuh:"$Amount"}}});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$sum:1}}});
db.cust_table.aggregate({$group:{_id:"$Cust_Name","total":{$addToSet:"$Amount"}}});
db.cust_table.createIndex({'Item_id':1})
db.cust_table.createIndex({'Item_id':2})
db.cust_table.createIndex({'Item_id':4})
db.cust_table.getIndexes()
db.cust_table.dropIndex({'Item_id':4})
db.cust_table.dropIndex({'Item_id':1})
db.cust_table.getIndexes()
```

Output :

```
test> switched to db customer
```

```
customer> DeprecationWarning: Collection.insert() is deprecated. Use insertOne,
insertMany, or bulkWrite.
```

```
{acknowledged: true,
  insertedIds: { '0': ObjectId('6904f6d9830bb476face5f4f') }}
```

```
customer> {acknowledged: true,
  insertedIds: { '0': ObjectId('6904f6d9830bb476face5f50') }}
```

```
customer> { acknowledged: true,
  insertedIds: { '0': ObjectId('6904f6d9830bb476face5f51') }}
```

```
customer> {acknowledged: true,
  insertedIds: { '0': ObjectId('6904f6d9830bb476face5f52') }}
```

```
customer> {acknowledged: true,
  insertedIds: { '0': ObjectId('6904f6d9830bb476face5f53') }}
```

```
customer> [{ _id: 'Shivam', total: 10 },
  { _id: 'Mohan', total: 100 },
  { _id: 'Ram', total: 90 }]
```

```
customer> [{ _id: 'Ram', total: 45 },
  { _id: 'Mohan', total: 50 },
  { _id: 'Shivam', total: 10 }]
```

```
customer> [{ _id: 'Shivam', total: 10 },
  { _id: 'Mohan', total: 40 },
  { _id: 'Ram', total: 40 }]
```

```
customer> [{ _id: 'Ram', total: 50 },
  { _id: 'Mohan', total: 60 },
  { _id: 'Shivam', total: 10 }]
```

```
customer> [ { _id: 'Shivam', total: 10 },
  { _id: 'Mohan', total: 40 },
  { _id: 'Ram', total: 40 }]
```

```
customer> [{ _id: 'Shivam', total: 10 },
  { _id: 'Mohan', total: 60 },
```

```

    { _id: 'Ram', total: 50 } ]
customer> Uncaught
MongoServerError[Location15952]: unknown group operator '$psuh'
customer> [
  { _id: 'Shivam', total: 1 },
  { _id: 'Mohan', total: 2 },
  { _id: 'Ram', total: 2 }
]
customer> [
  { _id: 'Shivam', total: [ 10 ] },
  { _id: 'Mohan', total: [ 40, 60 ] },
  { _id: 'Ram', total: [ 40, 50 ] }
]
customer> Item_id_1
customer> Item_id_2
customer> Item_id_4
customer> [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { Item_id: 1 }, name: 'Item_id_1' },
  { v: 2, key: { Item_id: 2 }, name: 'Item_id_2' },
  { v: 2, key: { Item_id: 4 }, name: 'Item_id_4' }
]
customer> { nIndexesWas: 4, ok: 1 }
customer> { nIndexesWas: 3, ok: 1 }
customer> [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { Item_id: 2 }, name: 'Item_id_2' }
]

```